

The NASA Astrobiology Institute (NAI) was established in 1998 by the National Aeronautics and Space Administration (NASA) "to develop the field of astrobiology and provide a scientific framework for flight missions." In December 2019 the institute's activities were suspended.

The NAI is a virtual, distributed organization that integrates astrobiology research and training programs in concert with the national and international science communities.

History

Although NASA had explored the idea of forming an astrobiology institute in the past, when the Viking biological experiments returned negative results for life on Mars, the public lost interest and federal funds for exobiology dried up. In 1996, the announcement of possible traces of ancient life in the Allan Hills 84001 meteorite from Mars led to new interest in the subject. At the same time, NASA developed the Origins Program, broadening its reach from exobiology to astrobiology, the study of the origin, evolution, distribution, and future of life in the universe.

In 1998, \$9 million was set aside to fund the NASA Astrobiology Institute (NAI), an interdisciplinary research effort using the expertise of different scientific research institutions and universities from across the country, centrally linked to Ames Research Center in Mountain View, California. Gerald Soffen former Project Scientist with the Viking program, helped coordinate the new institute. In May, NASA selected eleven science teams, each with a Principal Investigator (PI). NAI was established in July with Scott Hubbard as interim Director. Nobel laureate Baruch S. Blumberg was appointed the first Director of the institute, and served from May 15, 1999 – October 14, 2002.

Program

The NASA Astrobiology Program includes the NAI as one of four components, including the Exobiology and Evolutionary Biology Program; the Astrobiology Science and Technology

Instrument Development (ASTID) Program; and the Astrobiology Science and Technology for Exploring Planets (ASTEP) Program. Program budgets for fiscal year 2008 were as follows: NAI, \$16 million; Grants for the Exobiology and Evolutionary Biology Program, \$11 million; ASTID, \$9 million; ASTEP, \$5 million.

Teams

As of 2018, the NAI has 10 teams including about 600 researchers distributed across ~100 institutions. It also has 13 international partner organizations. Some past and present teams are:

International partners

NAI has partnership program with other international astrobiology organizations to provide collaborative opportunities for its researchers within the global science community.

Associate Partners

Spain Astrobiology Center (CAB) at the Instituto Nacional de Técnica Aeroespacial, Madrid, Spain

Australian Centre for Astrobiology (ACA) at the University of New South Wales

Affiliate Partners

Astrobiology Society of Britain (ASB)

Canadian Astrobiology Network (CAN) at Centre for Planetary Science and Exploration (CPSX), at the University of Western Ontario

European Exo/Astrobiology Network Association (EANA)

Helmholtz Alliance: Planetary Evolution and Life

Instituto de Astrobiología Colombia (IAC)

Japan AstroBiology Consortium (JABC), a partnership of the Earth-Life Science Institute and the National Institutes of Natural Sciences

Nordic Network of Astrobiology

Russian Astrobiology Center (RAC)

Sociedad Mexicana de Astrobiología (SOMA)

Société Française d'Exobiologie (SFE) [1]

UK Centre for Astrobiology at The University of Edinburgh

University of São Paulo (USP)

Research

Selected, significant topics of interdisciplinary research by NAI as of 2008:

Comets in space and in the laboratory

Discovery of the "rare biosphere"

Early habitability of Earth

Early wet Mars

Exoplanet discovery and analysis

Life without the Sun

Metal isotope tracers of environment and biology

Methane on Mars

Microbial mat ecology

Modeling exoplanet biospheres

Origins of life

Snowball Earth

Sub-seafloor life

The rise of oxygen and Earth's "middle age"

References

Further reading

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